

Question 1 (2 points)

Consider the following languages

$$L_1 = \{a^n b^n a^m \mid n, m \geq 0\}$$

$$L_2 = \{a^n b^m a^n \mid n, m \geq 0\}$$

Which of the following languages is DCFL?

- A. $\overline{L_1}$
- B. $L_1 \cap L_2$
- C. $L_1 \cup L_2$
- D. $L_1 \cap \overline{L_2}$

Option A is correct. L_1 and L_2 are both DCFLs. DCFLs are closed under complementation, but not under intersection and union.

[$L_1 \cap L_2$ and $L_1 \cap \overline{L_2}$ are not context-free languages. We can show that these languages do not satisfy the pumping lemma for context-free languages. On the other hand, $L_1 \cup L_2$ is context-free but not deterministic context-free.]

Question 2 (2 points)

Which of the following languages is not necessarily context-free?

- A. Complement of a DCFL
- B. Intersection of a DCFL and a regular language
- C. Union of two DCFLs
- D. Intersection of two DCFLs

Option D is correct. DCFLs are closed under complementation and hence the complement of a DCFL is also DCFL and hence context-free. Intersection of a DCFL and a regular language is also a DCFL and hence context-free. Since context-free languages are closed under union, therefore the union of two DCFLs is also context-free. But CFLs as well as DCFLs are not closed under intersection. Therefore intersection of two DCFLs may not be context-free.

Question 3 (2 points)

Which of the following languages is necessarily deterministic context-free?

- A. Complement of a DCFL
- B. Intersection of a DCFL and a regular language
- C. Union of two DCFLs
- D. Intersection of two DCFLs

Both Option A and B are correct. This question will be reevaluated.

DCFLs are not closed under union and intersection but they are closed under complementation and intersection with a regular language.

Question 4 (2 points)

Let L_1 and L_2 be two languages such that $L_1 \subseteq L_2$. Consider the following statements.

- S_1 : If L_1 is DCFL then L_2 is also DCFL
- S_2 : If L_2 is DCFL then L_1 is also DCFL
- S_3 : Either both L_1 and L_2 are DCFL or neither L_1 nor L_2 are DCFL

Which of the following is correct?

-
- A. S_1 is true
 - B. S_2 is true
 - C. S_3 is true
 - D. None of S_1, S_2, S_3 is true**

Option D is correct. The subset of a DCFL is not necessarily a DCFL. Similarly the superset of a DCFL is also not necessarily a DCFL. Eg.- the non-context-free language $\{a^n b^n c^n\}$ is a subset of the DCFL $\{a^n b^n c^m\}$. Similarly the non-context-free language $\{a^n b^m c^p \mid m \neq n \neq p\}$ is a superset of the DCFL $\{a^n b^m \mid m \neq n \neq 0\}$ (considering $p = 0$).

Question 5 (2 points)

Let L_1 be a context-free language, L_2 be a DCFL. Which of the following is not true?

- A. $\overline{L_1} \cup L_2$ is necessarily context-free**
- B. $L_1 \cup \overline{L_2}$ is necessarily context-free
- C. $L_1 \cdot \overline{L_2}$ is necessarily context-free
- D. $L_1 \cup L_2$ is necessarily context-free

Option A is correct. Since DCFLs are closed under complementation, therefore $\overline{L_2}$ is DCFL (hence, context-free). Now since context-free languages are closed under union and concatenation, therefore $L_1 \cup L_2$, $L_1 \cdot \overline{L_2}$, $L_1 \cup \overline{L_2}$ are all context-free. On the other hand, context-free languages are not closed under complementation and hence $\overline{L_1}$ is not necessarily context-free and therefore $\overline{L_1} \cup L_2$ is also not necessarily context-free.

Question 6 (2 points)

Which of the following statements is not true about a deterministic Turing Machine?

- A. Equivalent to a nondeterministic Turing Machine in terms of computability
- B. Can go to multiple configurations from a given configuration**
- C. Can have multiple accept configurations
- D. Accepts a decidable language

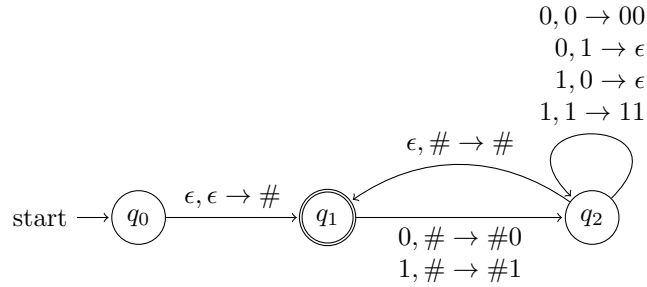
Option B is true. A deterministic Turing Machine can only move to a single configuration at once. (A non-deterministic Turing Machine can move to multiple configurations at once.)

Question 7 (2 points)

Which of the following is a deterministic context-free language?

- A. $\{ww^R \mid w \in \{0,1\}^* \text{ and } w^R \text{ is the reverse of } w\}$
- B. $\{ww \mid w \in \{0,1\}^*\}$
- C. $\{w \mid w \in \{0,1\}^* \text{ and } w \text{ has equal number of 0's and 1's}\}$**
- D. $\{a^n b^n c^m \text{ or } a^n b^m c^n \mid m, n \text{ are natural numbers}\}$

Option C is correct. Option A is the language of all even-length palindromes, which is known to be not a DCFL. (Refer to section 13.1 in Week 6 Lecture Notes). Option B is not even a context-free language. (Refer to section 10.2 in Week 5 Lecture Notes). Option D is not a DCFL although it is the union of two DCFLs (DCFLs are not closed under union.) Option C is a DCFL and here is a DPDA (by final state) accepting this language.



The above machine moves to the state q_1 whenever the stack gets empty. From state q_1 , it pushes a 0/1 (the next input symbol) and moves to state q_2 . When in state q_2 , the machine pushes the next input symbol onto the stack if it matches the symbol at the stack top. Otherwise, it pops one symbol out of the stack. The machine remains in state q_2 as long as the stack is not empty and once it gets empty, it moves to state q_1 . q_1 is the accept state, that is an input is accepted if and only if at the end of reading the input, the stack is empty. It is easy to see why this DPDA accepts all (and only those) strings that have equal number of 0's and 1's.

Question 8 (2 points)

Which of the following statements is not true about a Turing Machine?

- A. Can move across the work tape in both directions
- B. Can write on the work tape
- C. Has a fixed (independent of the input) number of states
- D. Has a fixed (independent of the input) number of configurations**

Option D is correct. The tape contents and how much of the tape is being used by the Turing machine depends on the input. Hence, the number of possible configurations of a Turing machine is not fixed. It depends on the input.